# Extremal no-three-in-line subsets of a modular hyperbola in the Hall–Jackson–Sudbery–Wild window


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**Abstract**

Let $p$ be an odd prime, $c \in \mathbb{F}_p^*$, and let $H(c,p) = \{(x,y) \in \mathbb{Z}^2 : xy \equiv c \pmod p\}$ be the modular hyperbola. Hall, Jackson, Sudbery and Wild (1975) observed that the $2p \times 2p$ window $G(p) = [-\frac{p-1}{2}, \frac{3p-1}{2}] \times [0, 2p-1]$ contains exactly four points of each of the $p-1$ residue classes of $H(c,p)$, at the vertices of a $p \times p$ square, and that keeping three of the four (the twelve outer blocks $T_2$ of the $4 \times 4$ block subdivision of $G(p)$) yields $3(p-1)$ points with no three collinear. Kovács, Nagy and Szabó (2025/26) state that this deletion is optimal, i.e. that $H(c,p) \cap T_2$ is a largest no-three-in-line subset of $H(c,p) \cap G(p)$; the value $3(p-1)$ is therefore not new. We give what is, to our knowledge, the first complete proof of this statement, for every $c$, and we determine the complete extremal structure: we classify all lines carrying three or more points of $H(c,p) \cap G(p)$ (they have slope $\pm 1$, and their number is $\frac{3}{2}(p-1) - s$), we exhibit an explicit linear-programming certificate for the bound $3(p-1)$ (so even the fractional relaxation has value $3(p-1)$), and we show that the number of maximum no-three-in-line subsets is exactly $9^s$, where $s \in \{0,1,2\}$ is the number of quadratic residues among $c$ and $-c$. In particular the HJSW set is the unique maximum if and only if $p \equiv 1 \pmod 4$ and $c$ is a quadratic non-residue. The bound does not depend on the position of the window: for *every* $2p \times 2p$ box $[x_0, x_0+2p) \times [y_0, y_0+2p)$ the modular hyperbola contains at most $3(p-1)$ points in general position, and we determine the exact maximum and the number of maximum sets for every box (an "orbit lemma": a box separates 0 or 2 of the four partner pairs of each generic orbit of the Klein four-group generated by $(a,b) \mapsto (-b,-a)$ and $(a,b) \mapsto (b,a)$, and the maximum is $12n_2 + 10n_1 + 8n_0 + 6s$ in terms of the numbers $n_2, n_1, n_0$ of generic orbits with 2, 1, 0 split pairs of each kind); the HJSW box is among the boxes where $3(p-1)$ is attained. The classification also gives the no-four-in-line analogue: the largest no-four-in-line subset of $H(c,p) \cap G(p)$ has exactly $\frac{7}{2}(p-1) + s$ points, so the set $S_3$ of Kovács–Nagy–Szabó is optimal as well. For the union of two hyperbolae $H(1) \cup H(-1)$ in the HJSW box we prove $\alpha \leq 4(p-1) - 4m_8(p)$ by an explicit weighted line cover, where $m_8(p)$ (the number of eight-point diagonals) is positive for all $19 \leq p \leq 1500$ and equals $(1/12 + o(1))\,p$ (by Bombieri's estimate along a cubic), so that $\alpha \leq (11/3 + o(1))(p-1)$; to our knowledge this is the first bound below the trivial $4(p-1)$ for a union of two hyperbolae. The proofs are elementary and self-contained; independent programs check the statements about the HJSW box for all $p \leq 101$ (all $c$ for $p \leq 31$), the bound $3(p-1)$ for all boxes with $p \leq 61$, and the box formula and orbit lemma for all boxes with $p \leq 13$ and $p \leq 31$ respectively.


## 1 Introduction

The no-three-in-line problem asks for the largest subset of the $n \times n$ grid with no three collinear points; at most $2n$ points are possible, and the best general lower bound, $\frac{3}{2}n - o(n)$, is due to Hall, Jackson, Sudbery and Wild [1]: for $n = 2p$, $p$ prime, they place $3(p-1)$ points of a modular hyperbola $H(c,p)$ in the window $G(p)$. Their construction rests on two facts: every line meets $H(c,p)$ in at most two residue classes, and $G(p)$ contains exactly four points of each class, at the


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vertices of a $p \times p$ square; keeping the three copies outside the middle block of the $4 \times 4$ block subdivision of $G(p)$ leaves no three collinear.

Kovács, Nagy and Szabó [2] recently used the same window and its refinements for the no-$(k+1)$-in-line problem. In their setting the HJSW set is the case $|W| = 1$, $k = 2$ of a general randomised construction (all four copies of every class are taken, then a minimum set $R$ of points is deleted so that every line of slope $\pm 1$ retains at most $k$ points), and they state that for $|W| = 1$, $k = 2$ "Subsection 2.2 essentially details an optimal choice for the points $R$ to be removed to gain the largest no-3-in-line set $S_2(c, p)$ inside $S(W)$" [2, App. A]. The general estimate of [2, Rem. 3.4], $|R| \geq \frac{1}{2}X$ with $X = \sum_\ell (|\ell \cap S(W)| - k)^+$, gives for $|W| = 1$, $k = 2$ the bound $|R| \geq p - 1 - s$ (see Remark 9 below), i.e. $|S| \leq 3(p-1) + s$ for every no-three-in-line $S \subseteq H(c, p) \cap G(p)$: exact when neither $c$ nor $-c$ is a square modulo $p$, but one or two short of the truth otherwise. We are not aware of a proof of the exact statement in [2] or elsewhere; the purpose of this note is to give one, and to determine the whole extremal structure of the problem. To be explicit about what is new: the value $3(p-1)$ and the qualitative optimality of $S_2$ and $S_3$ are asserted in [2]; the proof, the classification of the rich lines, the LP certificate, the count $9^s$ with its uniqueness criterion, the exact value and count for every $2p \times 2p$ box, and the exact no-four-in-line value and count are, as far as we could determine, new.

**Context.** Modular hyperbolas $xy \equiv c \pmod{m}$ have a rich literature (see Shparlinski's survey [7]; for combinatorial-geometric questions about them, e.g. [8]); the general problem of selecting a largest subset in general position from a given point set is NP-hard and APX-hard [9], with general bounds in [10], whereas here the structure of the point set makes the problem solvable exactly. Variants of the no-three-in-line problem with exact structural answers include the extensible version [11] and the torus version [12]. For $k \geq 3$ points per line the asymptotic answer $kn$ was recently established [13], while for $k = 2$ the HJSW construction is still the best known general lower bound (see Green's Problem 72 [14]); the state of the art for exact $2n$-point configurations is documented in Flammenkamp's database [15] and in [16].

**Results.** Throughout, $p$ is an odd prime, $h = (p-1)/2$, $c \in \mathbb{F}_p^*$, and

$$H(c, p) = \{(x, y) \in \mathbb{Z}^2 : \ xy \equiv c \pmod{p}\}, \qquad G(p) = \{-h, \dots, 3h+1\} \times \{0, \dots, 2p-1\},$$

and $\mathcal{P} := H(c, p) \cap G(p)$. A set is *lawful* (no-three-in-line, in general position) if it contains no three collinear points. A line is *rich* if it contains at least three points of $\mathcal{P}$; $\mathcal{L}$ denotes the set of rich lines. Let $s = [\, c \text{ is a QR}\,] + [\, -c \text{ is a QR}\,] \in \{0, 1, 2\}$ (QR = quadratic residue modulo $p$).

**Theorem 1.** *(i) Every rich line has slope $+1$ or $-1$; there are exactly $\frac{3}{2}(p-1) - s$ rich lines, $p - 1$ of them carrying three points of $\mathcal{P}$ and $\frac{1}{2}(p-1) - s$ carrying four (Proposition 8 lists them); exactly $2s$ points of $\mathcal{P}$ lie on no rich line.*

*(ii) Every lawful subset of $\mathcal{P}$ has at most $3(p-1)$ points; the bound is attained. Even the fractional relaxation (maximise $\sum_q x_q$ subject to $\sum_{q \in \ell} x_q \leq 2$ for $\ell \in \mathcal{L}$, $0 \leq x_q \leq 1$) has optimum $3(p-1)$.*

*(iii) The number of lawful subsets of size $3(p-1)$ is exactly $9^s$. Every maximum set coincides with the HJSW set $H(c, p) \cap T_2$ outside the (at most four) classes $(a, \pm a)$ with $a^2 = \pm c$.*

**Corollary 2.** *If $p \equiv 3 \pmod{4}$ then $s = 1$ for every $c$: there are exactly $9$ maximum sets. If $p \equiv 1 \pmod{4}$ then $s = 2$ for $c$ a QR ($81$ maximum sets) and $s = 0$ for $c$ a non-residue: the HJSW set is the unique maximum. Thus $H(c, p) \cap T_2$ is the unique largest lawful subset of $H(c, p) \cap G(p)$ if and only if $p \equiv 1 \pmod{4}$ and $c$ is a quadratic non-residue.*

**Theorem 3.** *Let $x_0, y_0 \in \mathbb{Z}$ and $W = [x_0, x_0 + 2p) \times [y_0, y_0 + 2p)$. Every lawful subset of $H(c, p) \cap W$ has at most $3(p-1)$ points. More precisely, the maximum equals $12n_2 + 10n_1 + 8n_0 + 6s$, where $n_2, n_1, n_0$ are the numbers of generic orbits of the Klein four-group whose partner pairs are split* *by the box in the patterns* $\{(0,2),(2,0)\}$, $(1,1)$, $(0,0)$ *respectively* ($n_2+n_1+n_0=\frac{1}{4}(p-1-2s)$*; no other pattern occurs), and the number of maximum sets is* $144^{n_1}\,1296^{n_0}\,9^{s_1}6^{s_2}$*, where* $s_1$ *(resp.* $s_2$*) is the number of exceptional orbits whose pair is split (resp. shared),* $s_1+s_2=s$*. In particular* $3(p-1)$ *is attained iff* $n_1=n_0=0$*; the HJSW box has* $n_2=\frac{1}{4}(p-1-2s)$, $s_1=s$*. (Maximum sets are counted as subsets of* $\mathbb{Z}^2$*, not up to symmetry. The pattern of an orbit and the split/shared label of an exceptional pair depend only on the orbit and the box, not on the class chosen to name the orbit.)*

The proof of Theorem 1 is a double count over the rich lines: for a lawful $S$, $|S|\le Z+2|\mathcal{L}|$ where $Z$ is the number of points on no rich line, and the identity $Z+2|\mathcal{L}|=3(p-1)$ follows from the classification of the rich lines. The count is exact precisely because the $2s$ points that lie on no rich line are accounted separately; this is the correction to the averaging bound of [2] mentioned above. All rich lines lie inside orbits of a Klein four-group acting on the classes, and the equality analysis reduces to two small "gadgets" (Figure 1), which gives $9^s$. For a general box the rich lines still lie inside the orbits, so the maximum is a sum of orbit maxima; the orbit lemma (Section 5) says that a box splits 0 or 2 of the four partner pairs of a generic orbit, and the four possible patterns have maxima 8, 10, 12, 12 — whence the exact formula of Theorem 3. For boxes other than the HJSW box the maximum can be smaller than $3(p-1)$ (Table 1). The four-point rich lines are pairwise disjoint, which settles the no-four-in-line analogue as well (Corollary 15: the maximum is $\frac{7}{2}(p-1)+s$ in the HJSW box, attained by the set $S_3$ of [2]). Section 6 turns to unions of two hyperbolae: for $H(1)\cup H(-1)$ in the HJSW box, Theorem 17 gives $\alpha\le 4(p-1)-4m_8(p)$ by an explicit line cover built from the eight-point diagonals and the two reflections of the box, and Proposition 22 evaluates $m_8(p)=(1/12+o(1))p$; Corollary 12 shows that among all conics only the shifted hyperbolae can reach $3(p-1)$ in a $2p\times 2p$ box. Section 7 collects the computations that motivate the (unproved) expectation that unions of hyperbolae gain only $O(1)$ over $3(p-1)$.

## 2 Notation

Two points are *congruent* if their coordinates agree modulo $p$; a *class* is a congruence class, identified with an element of $\mathbb{F}_p\times\mathbb{F}_p$. Points of $H(c,p)$ have both coordinates prime to $p$, so the classes of $H(c,p)$ are the $p-1$ pairs $\kappa=(a,b)\in\mathbb{F}_p^*\times\mathbb{F}_p^*$ with $ab=c$. Since each coordinate range of $G(p)$ is an interval of $2p$ consecutive integers, it contains exactly two representatives of every residue class; hence every class $\kappa$ of $H(c,p)$ has exactly four points in $G(p)$ and $|\mathcal{P}|=4(p-1)$. We write $(x_\kappa,y_\kappa)$ for the *base copy* of $\kappa$, its unique point in $\{-h,\dots,h\}\times\{1,\dots,p-1\}$ (necessarily $x_\kappa\ne 0$), and

$$\kappa(r,s):=(x_\kappa+rp,\ y_\kappa+sp),\qquad r,s\in\{0,1\},$$

for its four points in $G(p)$. Following [1, 2] we sort the classes into four *types* according to the position of the base copy:

$$A:\ x_\kappa>0,\ y_\kappa>h;\qquad B:\ x_\kappa<0,\ y_\kappa>h;\qquad C:\ x_\kappa>0,\ y_\kappa\le h;\qquad D:\ x_\kappa<0,\ y_\kappa\le h.$$

(In the notation of [2, Def. 2.3]: $\kappa(0,0)\in A_{0,0}$, $B_{-1,0}$, $C_{0,0}$, $D_{-1,0}$ respectively; the HJSW set $T_2$ omits, for each type, the copy $A_{0,0}$, $B_{0,0}$, $C_{0,1}$, $D_{0,1}$, i.e. $\kappa(0,0)$ for $A$, $\kappa(1,0)$ for $B$, $\kappa(0,1)$ for $C$, $\kappa(1,1)$ for $D$; the union of these omitted blocks is the middle block $M$ of [2, §2.3]. We call this copy the *middle copy* of $\kappa$.) We write $n_A,n_B,n_C,n_D$ for the numbers of classes of each type, so $n_A+n_B+n_C+n_D=p-1$. Finally, for a class $\kappa$ put

$$d_\kappa:=x_\kappa-y_\kappa,\qquad e_\kappa:=x_\kappa+y_\kappa,\qquad D_\kappa:=\{x-y=d_\kappa\},\qquad E_\kappa:=\{x+y=e_\kappa+p\};$$

$D_\kappa$ (the *diagonal* of $\kappa$) contains $\kappa(0,0),\kappa(1,1)$ and $E_\kappa$ (the *antidiagonal*) contains $\kappa(0,1),\kappa(1,0)$. For a real number $x$ let $x^+=\max\{x,0\}$.

Three involutions act on the classes of $H(c,p)$:

$$\sigma(a,b)=(-b,-a),\qquad \tau(a,b)=(b,a),\qquad \nu(a,b)=(-a,-b)=\sigma\tau(a,b)=\tau\sigma(a,b),$$

so $V=\{1,\sigma,\tau,\nu\}$ is a Klein four-group acting on the $p-1$ classes. $\sigma$ has fixed classes iff $-c$ is a QR (then exactly two: $(a,-a)$ with $a^2=-c$), $\tau$ has fixed classes iff $c$ is a QR (two: $(a,a)$ with $a^2=c$), and $\nu$ has none. Put $f_\sigma:=\#\{\kappa:\sigma\kappa=\kappa\}\in\{0,2\}$, $f_\tau:=\#\{\kappa:\tau\kappa=\kappa\}\in\{0,2\}$; thus $f_\sigma+f_\tau=2s$. The $V$-orbits are of size 4 (*generic*: four distinct classes) or of size 2 (*exceptional*: $\{\kappa,\nu\kappa\}$ with $\kappa$ and $\nu\kappa$ both fixed by $\sigma$, or both fixed by $\tau$); there are exactly $s$ exceptional orbits.

## 3 Rich lines

**Lemma 4.** *Let $\ell$ be a line containing at least two points of $H(c,p)$. Then the points of $H(c,p)\cap\ell$ lie in at most two classes. More precisely, if $\ell$ contains a point of class $\kappa$, then the classes met by $\ell$ are: only $\kappa$ if $\ell$ has slope $0$ or $\infty$; among $\{\kappa,\sigma\kappa\}$ if $\ell$ has slope $+1$; among $\{\kappa,\tau\kappa\}$ if $\ell$ has slope $-1$.*

*Proof.* Let $(u,v)\in\mathbb{Z}^2$ be a primitive direction vector of $\ell$ and $(x_0,y_0)\in H(c,p)\cap\ell$. The lattice points of $\ell$ are $(x_0+tu,y_0+tv)$, $t\in\mathbb{Z}$, and such a point lies on $H(c,p)$ iff $uv\,t^2+(x_0v+y_0u)\,t\equiv 0 \pmod p$ (we used $x_0y_0\equiv c$). The class of the point depends only on $t \bmod p$. If $uv\not\equiv 0$, the congruence has at most two roots modulo $p$; if exactly one of $u,v$ is divisible by $p$ (both cannot be, as $\gcd(u,v)=1$), it reduces to $x_0v\,t\equiv 0$ or $y_0u\,t\equiv 0$ with a unit coefficient, so $t\equiv 0$: one class. For $(u,v)=(1,0),(0,1)$ this gives the claim for slopes $0,\infty$. For $(u,v)=(1,1)$ the roots are $t\equiv 0$ and $t\equiv-(x_0+y_0)$, the latter giving the point $(-y_0,-x_0)$ modulo $p$, of class $\sigma\kappa$. For $(u,v)=(1,-1)$ the roots are $t\equiv 0$ and $t\equiv y_0-x_0$, giving $(y_0,x_0)$, of class $\tau\kappa$. $\square$

**Lemma 5.** *Two distinct points of one class in $G(p)$ differ by $\pm(p,0)$, $\pm(0,p)$, $\pm(p,p)$ or $\pm(p,-p)$; the line through them has slope $0,\infty,+1$ or $-1$ and contains no third point of that class. The point $\kappa(r,s)$ lies on the slope-$(+1)$ line $x-y=d_\kappa+(r-s)p$ and on the slope-$(-1)$ line $x+y=e_\kappa+(r+s)p$. In particular $D_\kappa$ and $E_\kappa$ are the only lines of slope $\pm1$ carrying two points of $\kappa$.*

*Proof.* The four points of a class are the vertices of an axis-parallel square of side $p$; no three are collinear. The rest is a direct computation. $\square$

**Corollary 6.** *A rich line has slope $+1$ or $-1$. Consequently a set $S\subseteq\mathcal{P}$ is lawful iff $|S\cap\ell|\le 2$ for every $\ell\in\mathcal{L}$.*

*Proof.* By Lemma 4 three points of $\mathcal{P}$ on a line lie in at most two classes, so two of them are congruent, and by Lemma 5 the line has slope $0,\infty,\pm1$. A line of slope 0 or $\infty$ meets a single class (Lemma 4), which has at most two points on it (Lemma 5). $\square$

**Lemma 7** (position of the partner classes)**.** *For every class $\kappa$ the types of $\sigma\kappa$, $\tau\kappa$ and the shifts $d_{\sigma\kappa}-d_\kappa$, $e_{\tau\kappa}-e_\kappa$ are given by the following table:*

| *type of* $\kappa$ | *type of* $\sigma\kappa$ | $d_{\sigma\kappa}-d_\kappa$ | *type of* $\tau\kappa$ | $e_{\tau\kappa}-e_\kappa$ |
|---|---|---|---|---|
| $A$ | $A$ | $0$ | $D$ | $-p$ |
| $B$ | $C$ | $+p$ | $B$ | $0$ |
| $C$ | $B$ | $-p$ | $C$ | $0$ |
| $D$ | $D$ | $0$ | $A$ | $+p$ |

*In particular $\sigma$ preserves the types $A$, $D$ and interchanges $B$, $C$; $\tau$ preserves $B$, $C$ and interchanges $A$, $D$; $\nu$ interchanges $A\leftrightarrow D$ and $B\leftrightarrow C$; the fixed classes of $\sigma$ lie in $A\cup D$, those of $\tau$ in $B\cup C$; and $n_A=n_D$, $n_B=n_C$.*

*Proof.* Write $\kappa = (a, b)$ with base copy $(x_\kappa, y_\kappa)$, so $x_\kappa \equiv a$, $y_\kappa \equiv b$. The base copy of $\sigma\kappa = (-b, -a)$ has first coordinate $-y_\kappa$ if $y_\kappa \le h$ and $p - y_\kappa$ if $y_\kappa > h$, and second coordinate $p - x_\kappa$ if $x_\kappa > 0$ and $-x_\kappa$ if $x_\kappa < 0$ (these are the representatives of $-b$ in $[-h, h]$ and of $-a$ in $[1, p-1]$). Going through the four types: $\kappa \in A$: $(p - y_\kappa,\ p - x_\kappa)$, again of type $A$, difference $x_\kappa - y_\kappa$; $\kappa \in D$: $(-y_\kappa,\ -x_\kappa)$, type $D$, difference $x_\kappa - y_\kappa$; $\kappa \in C$: $(-y_\kappa,\ p - x_\kappa)$, type $B$, difference $x_\kappa - y_\kappa - p$; $\kappa \in B$: $(p - y_\kappa,\ -x_\kappa)$, type $C$, difference $x_\kappa - y_\kappa + p$. Similarly, the base copy of $\tau\kappa = (b, a)$ has first coordinate $y_\kappa$ if $y_\kappa \le h$ and $y_\kappa - p$ otherwise, and second coordinate $x_\kappa$ if $x_\kappa > 0$ and $x_\kappa + p$ otherwise: $\kappa \in B$: $(y_\kappa - p,\ x_\kappa + p)$, type $B$, sum $e_\kappa$; $\kappa \in C$: $(y_\kappa,\ x_\kappa)$, type $C$, sum $e_\kappa$; $\kappa \in A$: $(y_\kappa - p,\ x_\kappa)$, type $D$, sum $e_\kappa - p$; $\kappa \in D$: $(y_\kappa,\ x_\kappa + p)$, type $A$, sum $e_\kappa + p$. The remaining assertions follow. ☐

**Proposition 8.** *$\mathcal{L}$ consists of the following lines, and no others; for each we list its points in $\mathcal{P}$.*

*(a) For $\kappa \in A \cup D$ with $\sigma\kappa \ne \kappa$: the common diagonal $D_\kappa = D_{\sigma\kappa}$, with the four points $\kappa(0,0), \kappa(1,1), \sigma\kappa(0,0), \sigma\kappa(1,1)$.*

*(b) For $\kappa \in B \cup C$ with $\tau\kappa \ne \kappa$: the common antidiagonal $E_\kappa = E_{\tau\kappa}$, with the four points $\kappa(0,1), \kappa(1,0), \tau\kappa(0,1), \tau\kappa(1,0)$.*

*(c) For every $\kappa \in C$: $D_\kappa$ with the three points $\kappa(0,0), \kappa(1,1), \sigma\kappa(1,0)$ $(\sigma\kappa \in B)$; for every $\kappa \in B$: $D_\kappa$ with the three points $\kappa(0,0), \kappa(1,1), \sigma\kappa(0,1)$ $(\sigma\kappa \in C)$.*

*(d) For every $\kappa \in A$: $E_\kappa$ with the three points $\kappa(0,1), \kappa(1,0), \tau\kappa(1,1)$ $(\tau\kappa \in D)$; for every $\kappa \in D$: $E_\kappa$ with the three points $\kappa(0,1), \kappa(1,0), \tau\kappa(0,0)$ $(\tau\kappa \in A)$.*

*The lines listed in (a)–(d) are pairwise distinct (apart from the identifications $D_\kappa = D_{\sigma\kappa}$ in (a) and $E_\kappa = E_{\tau\kappa}$ in (b) that are built into the list), and each of them lies inside a single $V$-orbit of classes. Consequently*

$$|\mathcal{L}| = \frac{n_A + n_D - f_\sigma}{2} + \frac{n_B + n_C - f_\tau}{2} + (n_A + n_D) + (n_B + n_C) = \frac{3}{2}(p-1) - s, \tag{1}$$

*with $p - 1$ three-point lines and $\frac{1}{2}(p-1) - s$ four-point lines. Moreover, the points of $\mathcal{P}$ lying on* no *rich line are exactly $\kappa(1,1)$ for $\sigma$-fixed $\kappa \in A$, $\kappa(0,0)$ for $\sigma$-fixed $\kappa \in D$, $\kappa(0,1)$ for $\tau$-fixed $\kappa \in B$ and $\kappa(1,0)$ for $\tau$-fixed $\kappa \in C$; their number is*

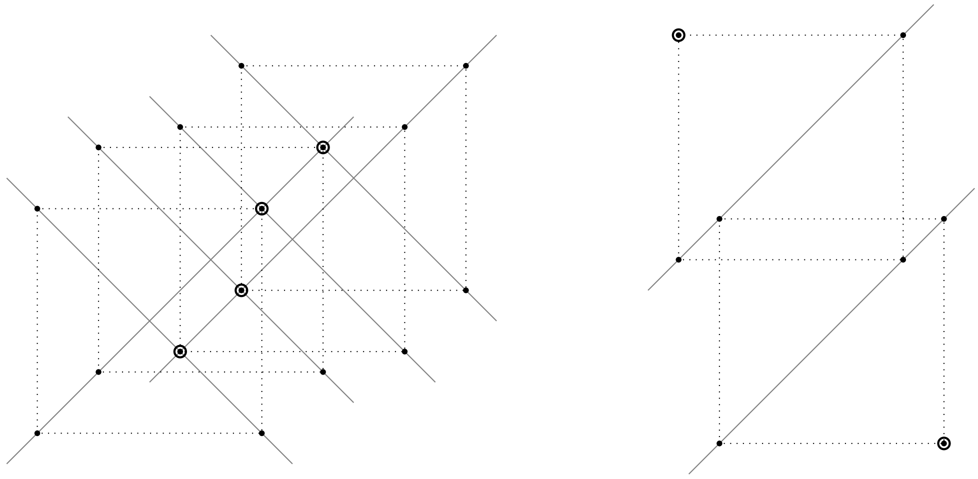
$$Z := \#\{q \in \mathcal{P} :\ q \text{ lies on no rich line}\} = f_\sigma + f_\tau = 2s. \tag{2}$$

*The points lying on* two *rich lines are exactly the middle copies of the classes $\kappa$ whose partner ($\sigma\kappa$ for $\kappa \in A \cup D$, $\tau\kappa$ for $\kappa \in B \cup C$) is different from $\kappa$; no point lies on three rich lines.*

*Proof.* Let $\ell \in \mathcal{L}$. By Corollary 6 its slope is $\pm 1$; say $+1$ (the case $-1$ is symmetric, with $\tau$, $e_\kappa$, $E_\kappa$ in place of $\sigma$, $d_\kappa$, $D_\kappa$). By Lemma 4 the points of $\ell \cap \mathcal{P}$ lie in two classes $\kappa, \sigma\kappa$ (a single class carries at most two points on a line), and one of the classes, say $\kappa$, has two points on $\ell$; by Lemma 5 these are $\kappa(0,0), \kappa(1,1)$ and $\ell = D_\kappa$, and $\sigma\kappa \ne \kappa$. The points of $\sigma\kappa$ on $D_\kappa$ are the $\sigma\kappa(r,s)$ with $d_{\sigma\kappa} + (r-s)p = d_\kappa$. By Lemma 7: for $\kappa \in A \cup D$, $r = s$, giving (a); for $\kappa \in C$, $r - s = 1$, giving $\sigma\kappa(1,0)$; for $\kappa \in B$, $r - s = -1$, giving $\sigma\kappa(0,1)$; this is (c). Conversely each listed line has at least three points.

Distinctness: if $D_\kappa = D_\lambda$ then this line of slope $+1$ meets both classes $\kappa, \lambda$, so $\lambda \in \{\kappa, \sigma\kappa\}$ by Lemma 4; and $D_\kappa = D_{\sigma\kappa}$ holds for $\kappa \in A \cup D$ (shift 0) but not for $\kappa \in B \cup C$ (shift $\pm p$). Likewise $E_\kappa = E_\lambda$ forces $\lambda \in \{\kappa, \tau\kappa\}$, and $E_\kappa = E_{\tau\kappa}$ holds exactly for $\kappa \in B \cup C$. A rich line meets only the classes $\kappa, \sigma\kappa$ (or $\kappa, \tau\kappa$), which lie in one $V$-orbit. Counting: (a) gives $(n_A + n_D - f_\sigma)/2$ lines, (b) $(n_B + n_C - f_\tau)/2$, (c) $n_B + n_C$, (d) $n_A + n_D$, which is (1).

For the incidences of a single point, take $\kappa \in A$. Its points $\kappa(0,1), \kappa(1,0)$ lie on $E_\kappa \in \mathcal{L}$ (d), and on no line of type (a)/(c) (their slope-(+1) lines $x - y = d_\kappa \pm p$ carry no second point



generic orbit ($p = 11$, $c = 1$, classes $(2,6), (5,9) \in A$, $(6,2), (9,5) \in D$): 16 points, 6 rich lines; the four circled middle copies lie on two rich lines each

exceptional orbit ($\tau$-fixed classes $(1,1) \in C$, $(10,10) \in B$): 8 points, 2 rich lines with 3 points each; the two circled points lie on no rich line

Figure 1: The two gadgets of the equality analysis, drawn to scale for $p = 11$, $c = 1$ (dotted squares: the four copies of a class; grey: rich lines). Generic orbit: the circled points are excluded and the other 12 are forced ($= T_2$). Exceptional orbit: the circled points are forced and each rich line keeps two of its three points, $3 \cdot 3 = 9$ choices.

of $\kappa$ and, by the table, no point of $\sigma\kappa$). $\kappa(1,1)$ lies on $D_\kappa$, which is rich iff $\sigma\kappa \neq \kappa$, and on $x + y = e_\kappa + 2p = e_{\tau\kappa} + 3p$, which carries no point of $\tau\kappa$: so $\kappa(1,1)$ is on one rich line if $\sigma\kappa \neq \kappa$ and on none otherwise. $\kappa(0,0)$ lies on $D_\kappa$ and on $x + y = e_\kappa = e_{\tau\kappa} + p$, i.e. on $E_{\tau\kappa} \in \mathcal{L}$: two rich lines if $\sigma\kappa \neq \kappa$, one otherwise. The types $D$, $B$, $C$ are analogous (for $B$: $\kappa(1,0) \in D_{\sigma\kappa}$ since $d_\kappa + p = d_{\sigma\kappa}$, while $\kappa(0,1)$ lies on $x - y = d_\kappa - p = d_{\sigma\kappa} - 2p$, which carries no point of $\sigma\kappa$; for $C$ the roles of $(1,0), (0,1)$ are exchanged; for $D$: $\kappa(1,1) \in E_{\tau\kappa}$ since $e_\kappa + 2p = e_{\tau\kappa} + p$, and $\kappa(0,0)$ lies on $x + y = e_\kappa = e_{\tau\kappa} - p$). This gives (2) and the last two assertions. $\square$

## 4 Proof of Theorem 1 and Corollary 2

Part (i) is Corollary 6 and Proposition 8.

*Upper bound.* Let $S \subseteq \mathcal{P}$ be lawful. For $q \in \mathcal{P}$ let $m(q)$ be the number of rich lines through $q$. Every rich line carries at most two points of $S$ (Corollary 6), hence

$$\begin{aligned} |S| \;&=\; \#\{q \in S : m(q) = 0\} + \#\{q \in S : m(q) \geq 1\} \;\leq\; Z + \sum_{q \in S} m(q) - \sum_{q \in S} (m(q) - 1)^+ \\ &=\; Z + \sum_{\ell \in \mathcal{L}} |S \cap \ell| - \sum_{q \in S} (m(q) - 1)^+ \;\leq\; Z + 2|\mathcal{L}|, \end{aligned} \tag{3}$$

and by (1), (2), $Z + 2|\mathcal{L}| = 2s + 3(p-1) - 2s = 3(p-1)$. The HJSW set attains the bound [1]; the equality analysis below exhibits all sets attaining it, so attainment does not depend on [1].

*LP certificate.* The same count is valid for fractional $x \in [0,1]^{\mathcal{P}}$ with $\sum_{q \in \ell} x_q \leq 2$ ($\ell \in \mathcal{L}$): $\sum_q x_q \leq \sum_{q : m(q) = 0} x_q + \sum_{\ell \in \mathcal{L}} \sum_{q \in \ell} x_q \leq Z + 2|\mathcal{L}|$. In LP-duality terms the dual solution assigns weight 1 to every rich line and weight 1 to the bound $x_q \leq 1$ of each of the $Z$ points on no rich line. This proves (ii). (The statement concerns the optimum value; we do not claim that the polytope of the relaxation is integral.)

*Equality.* $|S| = 3(p-1)$ holds iff all inequalities in (3) are equalities, i.e. iff (I) $S$ contains all $Z$ points lying on no rich line; (II) $|S \cap \ell| = 2$ for every $\ell \in \mathcal{L}$; (III) no point of $S$ lies on two rich lines. Since every rich line lies inside one $V$-orbit of classes (Proposition 8), conditions (I)–(III) split into independent conditions on the orbits, and the number of maximum sets is the product of the numbers of admissible choices in the orbits. In a generic orbit $\{\kappa, \sigma\kappa, \tau\kappa, \nu\kappa\}$ all four classes have distinct partners; the six rich lines meeting the orbit (two of type (a) or (b), four of type (c) or (d)) meet no other orbit, its four middle copies have $m = 2$ and are excluded by (III), and each of the six lines then retains exactly two points, both with $m = 1$, which are forced into $S$ by (II): on a generic orbit $S$ coincides with $H(c, p) \cap T_2$, and there is exactly one admissible choice. An exceptional orbit $\{\kappa, \nu\kappa\}$ ($\kappa$ and $\nu\kappa$ fixed by $\sigma$, or both fixed by $\tau$) meets exactly two rich lines, each with three points of $m = 1$, and contains two points with $m = 0$: conditions (I)–(III) allow $\binom{3}{2}\binom{3}{2} = 9$ choices, and all of them are lawful by Corollary 6. There are $s$ exceptional orbits, whence $9^s$ maximum sets, all agreeing with the HJSW set outside the exceptional orbits, i.e. outside the classes $(a, \pm a)$ with $a^2 = \pm c$. This proves (iii).

*Corollary 2.* $-1$ is a QR modulo $p$ iff $p \equiv 1 \pmod 4$. If $p \equiv 3 \pmod 4$, exactly one of $c, -c$ is a QR, so $s = 1$; if $p \equiv 1 \pmod 4$, $c$ and $-c$ have the same character, so $s \in \{0, 2\}$ according as $c$ is a non-residue or a residue. □

*Remark* 9. (a) In the language of [2, Constr. 3.2, Rem. 3.4] with $W = \{c\}$, $k = 2$ (so $S(W) = \mathcal{P}$ and $L'$ is the set of lines of slope $\pm 1$): $X_{2,p}(\{c\}) = \sum_\ell (|\ell \cap \mathcal{P}| - 2)^+ = (p-1)\cdot 1 + (\frac{1}{2}(p-1) - s)\cdot 2 = 2(p-1) - 2s$ by Theorem 1(i), so the general bound $|R| \geq \frac{1}{2}X$ gives $|R| \geq p - 1 - s$ and hence, for every lawful $S \subseteq \mathcal{P}$, $|S| \leq |\mathcal{P}| - |R| \leq 3(p-1) + s$ (a lawful $S$ is a trimming, by Corollary 6). The averaging loses exactly one point per exceptional orbit, because each such orbit contains two points on no rich line and no point on two rich lines; the count (3) keeps track of these points and is exact.

(b) The theorem says that no lawful set can use the middle block $M$ profitably: adding fourth copies to the HJSW set is impossible without removing at least as many points. Any improvement of the HJSW construction must therefore use points outside $H(c, p) \cap G(p)$ (another hyperbola, a larger window — by Theorem 3 not merely a shifted $2p \times 2p$ box — or non-hyperbola points).

(c) The proof uses only: at most two classes per line (Lemma 4), the square arrangement of the four copies, and the position of the partner classes (Lemma 7). It does not use $p \geq 7$; $p = 3$ and $p = 5$ are covered as well.

# 5 All $2p \times 2p$ boxes

Let $W = [x_0, x_0 + 2p) \times [y_0, y_0 + 2p)$ with $x_0, y_0 \in \mathbb{Z}$ (only $x_0, y_0$ modulo $p$ matter), $\mathcal{P}_W = H(c, p) \cap W$, and let $\rho_x(u) \in [x_0, x_0 + p)$, $\rho_y(u) \in [y_0, y_0 + p)$ denote the representatives of a residue $u$. Every class $\kappa = (a, b)$ of $H(c, p)$ again has exactly four points $\kappa(r, s) = (\rho_x(a) + rp, \rho_y(b) + sp)$, $r, s \in \{0, 1\}$, in $W$; put $d_\kappa = \rho_x(a) - \rho_y(b)$, $e_\kappa = \rho_x(a) + \rho_y(b)$, $D_\kappa = \{x - y = d_\kappa\}$, $E_\kappa = \{x + y = e_\kappa + p\}$. Lemmas 4, 5 and Corollary 6 hold verbatim (they do not use the position of the box), so a rich line is $D_\kappa$ for a class with $\sigma\kappa \neq \kappa$ or $E_\kappa$ for a class with $\tau\kappa \neq \kappa$, and its points are the two copies $\kappa(0, 0), \kappa(1, 1)$ (resp. $\kappa(0, 1), \kappa(1, 0)$) together with the copies of the partner class on the same line. Since $d_\kappa \equiv d_{\sigma\kappa} \pmod p$ and both lie in an interval of length $2p - 2$, $d_{\sigma\kappa} - d_\kappa \in \{-p, 0, p\}$; we call the $\sigma$-pair $\{\kappa, \sigma\kappa\}$ *shared* if $d_{\sigma\kappa} = d_\kappa$ (then $D_\kappa = D_{\sigma\kappa}$ carries the four points $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 0), \sigma\kappa(1, 1)$) and *split* otherwise (then $D_\kappa$ carries $\kappa(0, 0), \kappa(1, 1)$ and one copy of $\sigma\kappa$, and $D_{\sigma\kappa}$ carries $\sigma\kappa(0, 0), \sigma\kappa(1, 1)$ and one copy of $\kappa$). Likewise a $\tau$-pair is shared ($E_\kappa = E_{\tau\kappa}$, four points) or split (two lines with three points). In the HJSW box the $\sigma$-pairs in $A \cup D$ and the $\tau$-pairs in $B \cup C$ are shared and the others are split (Lemma 7); in the box $[0, 2p)^2$ all pairs are shared.

**Lemma 10** (orbit lemma)**.** *Let $O = \{\kappa, \sigma\kappa, \tau\kappa, \nu\kappa\}$ be a generic orbit, and let $e_O$ (resp. $f_O$) be the number of split $\sigma$-pairs (resp. $\tau$-pairs) among the two $\sigma$-pairs and two $\tau$-pairs of $O$. Then $e_O + f_O \in \{0, 2\}$; in particular $e_O + f_O \leq 2$.*

*Proof.* Write $\kappa = (a, b)$, $X_u = \rho_x(u)$, $X'_u = \rho_x(-u)$, $Y_u = \rho_y(u)$, $Y'_u = \rho_y(-u)$ for $u \in \{a, b\}$. The base copies are $P = (X_a, Y_b)$ of $\kappa$, $Q = (X'_b, Y'_a)$ of $\sigma\kappa$, $R = (X_b, Y_a)$ of $\tau\kappa$ and $T = (X'_a, Y'_b)$ of $\nu\kappa$; the $\sigma$-pairs are $\{P, Q\}$, $\{R, T\}$ and the $\tau$-pairs $\{P, R\}$, $\{Q, T\}$. With $d(x, y) = x - y$, $e(x, y) = x + y$, the four split conditions are $E_1 : d_P \neq d_Q$, $E_2 : d_T \neq d_R$, $F_1 : e_P \neq e_R$, $F_2 : e_Q \neq e_T$; each of the four differences is $\equiv 0 \pmod p$ and lies in $[-p, p]$, hence is one of $-p, 0, p$.

Let $m_x$ be the least multiple of $p$ exceeding $2x_0$ and $\mu_x = m_x - 2x_0 \in [1, p]$; similarly $m_y$, $\mu_y$. Since $X_u + X'_u \equiv 0 \pmod p$ and $X_u + X'_u \in [2x_0 + 1, 2x_0 + 2p - 3]$, we have $X_u + X'_u = m_x + \varepsilon_x(u)p$ with $\varepsilon_x(u) \in \{0, 1\}$, and explicitly $\varepsilon_x(u) = [X_u - x_0 > \mu_x]$; likewise $Y_u + Y'_u = m_y + \varepsilon_y(u)p$, $\varepsilon_y(u) = [Y_u - y_0 > \mu_y]$. Put $\Delta = \varepsilon_x(a) - \varepsilon_x(b)$, $\Gamma = \varepsilon_y(a) - \varepsilon_y(b)$, $A = \varepsilon_x(b) - \varepsilon_y(a)$, $B = \varepsilon_x(a) - \varepsilon_y(b)$, and define the integers $\theta, \eta$ by

$$(X_a + X_b) - (Y_a + Y_b) - (m_x - m_y) = \theta p, \qquad (X_b - X_a) - (Y_b - Y_a) = \eta p.$$

Substituting $X'_u = m_x + \varepsilon_x(u)p - X_u$, $Y'_u = m_y + \varepsilon_y(u)p - Y_u$ one finds

$$d_P - d_Q = (\theta - A)p, \quad d_T - d_R = (B - \theta)p, \quad e_R - e_P = \eta p, \quad e_Q - e_T = (\Gamma - \Delta - \eta)p,$$

so that $E_1 \Leftrightarrow \theta \neq A$, $E_2 \Leftrightarrow \theta \neq B$, $F_1 \Leftrightarrow \eta \neq 0$, $F_2 \Leftrightarrow \eta \neq \Gamma - \Delta$, and $\theta - A$, $\theta - B$, $\eta$, $\eta - (\Gamma - \Delta) \in \{-1, 0, 1\}$.

*Parity.* For $x \in \{-1, 0, 1\}$, $[x \neq 0] \equiv x \pmod 2$. Hence $E_1 + E_2 + F_1 + F_2 \equiv (\theta - A) + (\theta - B) + \eta + (\eta - \Gamma + \Delta) \equiv A + B + \Gamma - \Delta = 2\varepsilon_x(b) - 2\varepsilon_y(b) \equiv 0 \pmod 2$: the number of split pairs in $O$ is even.

*If $F_1$ holds, then $E_1$ and $E_2$ do not both hold.* Put $s = X_a - x_0$, $t = X_b - x_0 \in [0, p - 1]$ and $v = (x_0 - y_0) \bmod p$; since $Y_u \equiv X_u$, we have $Y_a - y_0 = s + v - c_a p$ and $Y_b - y_0 = t + v - c_b p$ with $c_a = [s + v \geq p]$, $c_b = [t + v \geq p]$, and therefore $\eta = c_b - c_a$. So $F_1$ means $c_a \neq c_b$; the substitution $a \leftrightarrow b$ exchanges $E_1 \leftrightarrow E_2$ and fixes $F_1$, so we may assume $c_a = 0$, $c_b = 1$, i.e. $s \leq p - 1 - v < t$. Since $\mu_x - \mu_y \equiv -2v \pmod p$ and $|\mu_x - \mu_y| \leq p - 1$, we have $\mu_x - \mu_y = -2v + kp$ with $k \in \{0, 1, 2\}$, and $\theta p = (2x_0 + s + t) - (2y_0 + s + t + 2v - p) - (\mu_x + 2x_0 - \mu_y - 2y_0) = p - 2v - (\mu_x - \mu_y) = (1 - k)p$, i.e. $\theta = 1 - k$. With $\varepsilon_x(a) = [s > \mu_x]$, $\varepsilon_x(b) = [t > \mu_x]$, $\varepsilon_y(a) = [s + v > \mu_y]$, $\varepsilon_y(b) = [t + v - p > \mu_y]$ and $\mu_y = \mu_x + 2v - kp$:

- $k = 1$: $E_1 \Leftrightarrow [t > \mu_x] \neq [s > \mu_x + v - p]$ and $E_2 \Leftrightarrow [s > \mu_x] \neq [t > \mu_x + v]$. If $s > \mu_x$ then $t > \mu_x$ and $s > \mu_x + v - p$, so $E_1$ fails. If $s \leq \mu_x$, $E_2$ forces $t > \mu_x + v$, so $\mu_x + v \leq p - 2$ and $[t > \mu_x] = 1$, and $E_1$ forces $s \leq \mu_x + v - p < 0$, impossible.
- $k = 0$: here $\mu_y = \mu_x + 2v \leq p$, so $\varepsilon_y(b) = 0$ and $E_2 \Leftrightarrow s \leq \mu_x$; then $\varepsilon_y(a) = [s > \mu_x + v] = 0$ and $E_1 \Leftrightarrow t \leq \mu_x$, but $t \geq p - v > p - 2v \geq \mu_x$.
- $k = 2$: here $\mu_y = \mu_x + 2v - 2p \geq 1$, so $\mu_x \geq 2p + 1 - 2v$ and $\varepsilon_y(a) = 1$; $E_1 \Leftrightarrow t > \mu_x$ and $E_2 \Leftrightarrow (s > \mu_x) \vee (t \leq \mu_x + v - p)$. Now $s > \mu_x$ would give $2p + 2 - 2v \leq s \leq p - 1 - v$, i.e. $v \geq p + 3$; so $s \leq \mu_x$ and $E_2$ needs $t \leq \mu_x + v - p < \mu_x$, contradicting $E_1$.

Consequently, if $F_1$ holds then $E_1 + E_2 + F_1 + F_2 \leq 3$, and if $F_1$ fails then again $E_1 + E_2 + F_1 + F_2 \leq 3$; by parity the sum is 0 or 2. $\square$

**Lemma 11** (gadget maxima)**.** *Let $O$ be an orbit and $\mathcal{P}_O \subseteq \mathcal{P}_W$ its 16 (or 8) points. Every rich line meeting $\mathcal{P}_O$ lies in $\mathcal{P}_O$, and the largest lawful subset of $\mathcal{P}_O$ has size and multiplicity as follows: exceptional orbit: 6 points, 9 sets if the pair is split and 6 if it is shared; generic orbit with $(e_O, f_O) = (0, 0)$: 8 points, $6^4 = 1296$ sets; $(1, 1)$: 10 points, 144 sets; $(0, 2)$ or $(2, 0)$: 12 points, one set.*

*Proof.* A rich line meets only a class and its $\sigma$- or $\tau$-partner (Lemma 4), hence lies in one orbit. *Exceptional orbit* $\{\kappa, \nu\kappa\}$, say $\sigma\kappa = \kappa$: the rich lines are $E_\kappa = E_{\nu\kappa}$ (four points) if the $\tau$-pair is shared, resp. $E_\kappa$, $E_{\nu\kappa}$ (three points each) if it is split; the remaining four (resp. two) points are free: $2+4=6$ points in $\binom{4}{2}=6$ ways, resp. $2+2+2=6$ points in $3\cdot 3=9$ ways. *Pattern* $(0,0)$: the two shared diagonals and the two shared antidiagonals are four disjoint four-point lines covering $\mathcal{P}_O$: $4\cdot 2=8$ points in $6^4$ ways. *Pattern* $(2,0)$ (and symmetrically $(0,2)$): the four diagonals $D_\lambda$ ($\lambda\in O$) are rich with three points each and the two antidiagonals $E_\kappa=E_{\tau\kappa}$, $E_{\sigma\kappa}=E_{\nu\kappa}$ have four points; every class $\lambda$ has exactly one point of multiplicity $m=2$, namely its off-diagonal copy lying on $D_{\sigma\lambda}$ (which is also on the antidiagonal of $\lambda$). Each $D_\lambda$ carries $\lambda(0,0),\lambda(1,1)$ ($m=1$) and one $m=2$ point, each antidiagonal carries two $m=1$ and two $m=2$ points. By (3) restricted to $O$ (with $Z=0$), $|S\cap\mathcal{P}_O|\le 12$ with equality iff $S$ avoids the four $m=2$ points and takes both remaining points of each of the six lines — which is lawful; so the maximum is 12 and it is unique. (For the HJSW box this is the generic orbit of Section 4.) *Pattern* $(1,1)$: the split $\sigma$-pair and the split $\tau$-pair share exactly one class $\kappa$ (any $\sigma$-edge and any $\tau$-edge of the four-cycle $\kappa-\tau\kappa-\nu\kappa-\sigma\kappa-\kappa$ meet in one vertex). The six rich lines are

$$\begin{array}{lllll} A:=D_\kappa, & B:=E_\kappa, & A':=D_{\sigma\kappa}, & B':=E_{\tau\kappa} & \text{(three points each)},\\ A'':=D_{\tau\kappa}=D_{\nu\kappa}, & B'':=E_{\sigma\kappa}=E_{\nu\kappa} & & & \text{(four points each)}. \end{array}$$

Write $\kappa_A,\kappa'_A$ for the two copies of $\kappa$ on $A$ (these are $\kappa(0,0),\kappa(1,1)$) and $\kappa_B,\kappa'_B$ for the two on $B$; the third point of $A$ is a copy $\bar\sigma$ of $\sigma\kappa$ off its own diagonal, hence on $B''$; the third point of $B$ is a copy $\bar\tau$ of $\tau\kappa$ off its own antidiagonal, hence on $A''$; the third point of $A'$ is a copy of $\kappa$ off $A$, i.e. one of $\kappa_B,\kappa'_B$ — name it $\kappa_B$; the third point of $B'$ is one of $\kappa_A,\kappa'_A$ — name it $\kappa_A$. No other incidences occur (a rich line of slope $+1$ through a copy of $\kappa$ is $A$ or $A'$, of slope $-1$ it is $B$ or $B'$, and $A\cap B=\emptyset$ because a common lattice point would have $x=x_\kappa+p/2$). Hence $m=2$ exactly for the four points $\kappa_A\in A\cap B'$, $\kappa_B\in B\cap A'$, $\bar\sigma\in A\cap B''$, $\bar\tau\in B\cap A''$, and $m=1$ for the other twelve: $A$ and $B$ consist of one $m=1$ point and two $m=2$ points, $A'$ and $B'$ of two $m=1$ points and one $m=2$ point, $A''$ and $B''$ of three $m=1$ points and one $m=2$ point. For lawful $S\subseteq\mathcal{P}_O$ let $t$ be the number of $m=2$ points in $S$; since every point of $\mathcal{P}_O$ lies on a rich line, $|S|=\sum_\ell|S\cap\ell|-t$ over the six lines, each term being $\le 2$. If $t=0$ then $|S\cap A|,|S\cap B|\le 1$ and $|S|\le 1+1+2+2+2+2=10$; if $t=1$, the $m=2$ point lies on exactly one of $A,B$, the other of the two carries at most one point of $S$, and $|S|\le 11-1=10$; if $t\ge 2$ then $|S|\le 12-t\le 10$. So the maximum is at most 10, and a set of size 10 makes every inequality used tight. Counting these sets by $t$ (every set listed is lawful, since each rich line receives at most two of its points):

$t=0$: $S$ contains $\kappa'_A,\kappa'_B$, both $m=1$ points of $A'$ and both of $B'$ (forced), and two of the three $m=1$ points of $A''$ and of $B''$: $3\cdot 3=9$ sets.

$t=1$: the $m=2$ point $q$ is one of $\kappa_A,\bar\sigma\in A$ or $\kappa_B,\bar\tau\in B$; say $q\in A$ (the case $q\in B$ is symmetric). Then $S\cap A=\{q,\kappa'_A\}$ and $S\cap B=\{\kappa'_B\}$; the second line through $q$ receives $q$ and one of its $m=1$ points, and each of the three remaining lines receives two of its $m=1$ points. For $q=\kappa_A$: $B'$ in 2 ways, $A'$ forced, $A''$ and $B''$ in $3\cdot 3$ ways — 18 sets; for $q=\bar\sigma$: $B''$ in 3 ways, $A'$ and $B'$ forced, $A''$ in 3 ways — 9 sets. With the symmetric case $q\in B$: $2\cdot(18+9)=54$ sets.

$t=2$: the two $m=2$ points cannot both lie on $A$ (then $|S\cap B|\le 1$ and $|S|\le 11-2=9$), nor both on $B$; so one is on $A$ and one on $B$, every line receives exactly two points, and $S\cap A=\{q,\kappa'_A\}$, $S\cap B=\{q',\kappa'_B\}$. For $\{q,q'\}=\{\kappa_A,\kappa_B\}$: $B'$ and $A'$ each receive their $m=2$ point and one of two $m=1$ points $(2\cdot 2)$, $A''$, $B''$ two of three $(3\cdot 3)$: 36 sets; for $\{\kappa_A,\bar\tau\}$: $B'$ 2 ways, $A''$ ($\ni\bar\tau$) 3 ways, $A'$ forced, $B''$ 3 ways: 18; for $\{\bar\sigma,\kappa_B\}$: symmetrically 18; for $\{\bar\sigma,\bar\tau\}$: $B''$ 3, $A''$ 3, $A'$, $B'$ forced: 9. Total $36+18+18+9=81$.

$t\ge 3$: $|S|\le 12-t\le 9$: no maximum set.

Altogether $9+54+81=144$ maximum sets of size 10. (An independent brute-force enumeration of the 16-point gadgets, `slack/gadget11_check.py`, confirms the line structure and the counts $9,54,81$ for every $(1,1)$-orbit encountered in all boxes for $p\le 13$ and in samples of boxes for $p\le 31$.) □

*Proof of Theorem 3.* By Lemma 11 the rich lines of $\mathcal{P}_W$ lie inside the orbits, so a set is lawful iff its intersection with every orbit is lawful, and the maximum and the number of maximum sets are the sum, resp. the product, of the orbit values. By Lemma 10 every generic orbit has pattern $(0,0)$, $(1,1)$, $(0,2)$ or $(2,0)$, and Lemma 11 gives the formula. Since $n_2+n_1+n_0 = \frac{1}{4}(p-1-2s)$, the maximum is $12n_2+10n_1+8n_0+6s \le 12(n_2+n_1+n_0)+6s = 3(p-1)$, with equality iff $n_1 = n_0 = 0$. For the HJSW box, Lemma 7 gives $(e,f) = (0,2)$ for the $A/D$-orbits and $(2,0)$ for the $B/C$-orbits, and the exceptional pairs are split; this recovers Theorem 1(ii),(iii). ☐

**Corollary 12** (all conics)**.** *Let $C$ be a non-degenerate conic over $\mathbb{F}_p$ and $c \ge 2$. If the projective closure of $C$ contains both axial points at infinity* (1:0:0) *and* (0:1:0)*, then $C$ is a shifted modular hyperbola $(x+e)(y+d) = c'$, and for $c = 2$ every lawful subset of its lift into a $2p \times 2p$ box has at most $3(p-1)$ points (Theorem 3 after the shift). Otherwise every lawful subset of the lift of $C$ into a box of side $cp$ has at most $c(p+3)$ points; for $c = 2$ and $p \ge 11$ this is less than $3(p-1)$. Hence for $p \ge 11$ no conic admits more than $3(p-1)$ lawful points in a $2p \times 2p$ box, and only the shifted modular hyperbolae can attain $3(p-1)$ (they do so exactly in the boxes described by Theorem 3, e.g. the HJSW box; a shifted hyperbola in a box where $n_1+n_0 > 0$ attains less, and a conic that is $\mathbb{F}_p$-affinely equivalent to a hyperbola without being one, e.g. $(x-y)y = 1$, is not covered by the shift and satisfies only the bound $c(p+3)$ — for $p = 7$ its lift into $G(7)$ has maximum $16 < 18$).*

*Proof.* A conic through (1:0:0) and (0:1:0) has no $x^2$ and no $y^2$ term, so its equation is $xy+dx+ey+f = 0$, i.e. $(x+e)(y+d) = de-f$, a shifted hyperbola (non-degenerate iff $de \ne f$). If, say, $(1{:}0{:}0) \notin C$, then $C$ has an $x^2$ term, so every horizontal line $y = b$ meets $C$ in at most two points, and the number of residues $b$ with $C \cap \{y = b\} \ne \emptyset$ is at most $\lceil |C(\mathbb{F}_p)|/2\rceil + 1 \le (p+3)/2$ (only the $\le 2$ points with vertical tangent are alone on their line). In a box of side $cp$ the lift of $C$ therefore meets at most $c(p+3)/2$ rows, and a lawful set has at most two points per row. The symmetric argument applies if $(0{:}1{:}0) \notin C$. ☐

(A degenerate conic is a union of at most two lines, and the lift of an $\mathbb{F}_p$-line into a box of side $cp$ lies on $O(c\sqrt{p})$ integer lines, so it contains $O(c\sqrt{p})$ lawful points. Note that $\mathbb{F}_p$-affine equivalence is *not* a symmetry of the lifted problem: only integer translations of the box are. The observation that only the shifted hyperbolae contain both axial points at infinity was made independently by reasoning agents consulted by the author of record, who also supplied the $p = 7$ example.)

*Remark* 13 (algorithmic consequence). Selecting a largest subset in general position from an arbitrary point set is NP-hard [9]; for $H(c,p) \cap W$ Theorems 1 and 3 reduce it to bookkeeping: after listing the $p-1$ classes and their $V$-orbits, the maximum, the number of maximum sets and an explicit maximum set are obtained in $O(p)$ operations (the pattern of an orbit is read off from the split/shared status of its four pairs).

*Remark* 14. The double count of Section 4 gives, for every box, $|S| \le Z + 2|\mathcal{L}| = 2(p-1)+2s+2\sum_O(e_O+f_O) = 12n_2+12n_1+8n_0+6s$ (one checks $Z+2|\mathcal{L}| = 2(p-1)+2s+\#\{m=2\}$ and $\#\{m=2\} = 2\sum_O(e_O+f_O)$ exactly as in Section 4); it is off by exactly 2 on every $(1,1)$-orbit, where two rich three-point lines consist of one point of multiplicity one and two of multiplicity two.

**Corollary 15** (no-four-in-line)**.** *In every box $W$ the four-point rich lines are pairwise disjoint; hence the largest subset of $\mathcal{P}_W$ with no four collinear points has exactly $4(p-1) - N_{sh}(W)$ points, where $N_{sh}(W)$ denotes the number of shared pairs ($\sigma$-pairs and $\tau$-pairs together; equivalently, the number of four-point rich lines of $\mathcal{P}_W$), and there are $4^{N_{sh}(W)}$ such subsets. By the orbit lemma $N_{sh}(W) \ge 2 \cdot \#\{generic\ orbits\} = \frac{1}{2}(p-1) - s$, so the maximum is at most $\frac{7}{2}(p-1)+s$; in the HJSW box $N_{sh} = \frac{1}{2}(p-1) - s$ and the set $S_3 = H(c,p) \cap T_3$ of [2, Constr. 2.10] (with $|S_3| = \frac{7}{2}(p-1)+s$) is a largest no-four-in-line subset of $H(c,p) \cap G(p)$, as asserted in [2, App. A]. (No line carries five points of $\mathcal{P}_W$, so the no-five-in-line maximum is $|\mathcal{P}_W| = 4(p-1)$.)*

*Proof.* A four-point rich line is $D_\kappa$ for a shared $\sigma$-pair (points $\kappa(0,0), \kappa(1,1), \sigma\kappa(0,0), \sigma\kappa(1,1)$) or $E_\kappa$ for a shared $\tau$-pair (points with $r+s=1$). The slope-$(-1)$ line through a point $\kappa(r,r)$ is not of the second kind: if it were $E_\lambda$ then $\kappa \in \{\lambda, \tau\lambda\}$ by Lemma 4, so $E_\lambda = E_\kappa$, which contains only the copies $\kappa(0,1), \kappa(1,0)$ of $\kappa$; likewise the slope-$(+1)$ line through a point $\kappa(r, 1-r)$ is not of the first kind. Hence the four-point lines are pairwise disjoint, a no-four-in-line subset must miss at least one point of each of them, and deleting exactly one point from each is sufficient (all other lines carry at most three points). For the HJSW box, $N_{sh} = \frac{1}{2}(p-1) - s$ by Lemma 7, and $|S_3| = |S_2| + |M \cap E \cap H(c,p)| = 3(p-1) + \frac{1}{2}(p-1) + s$ because the classes fixed by $\sigma$ or $\tau$ lie in $E$ and the remaining classes split evenly between $E$ and $F$ [2, Obs. 2.9]. $\square$

For the box $[0, 2p)^2$ all pairs are shared ($n_2 = n_1 = 0$, $s_2 = s$): the maximum is $8n_0 + 6s = 2(p-1) + 2s$ with $1296^{n_0}6^s$ maximum sets. Table 1 shows the exact maxima for all boxes at $p = 11$, $c = 1$ (two generic orbits, $s = 1$): the values $22, 24, 26, 28, 30$ correspond to $(n_1, n_0) = (0,2), (1,1), (2,0)$ or $(0,1)$, $(1,0)$, $(0,0)$. The number of boxes attaining $3(p-1)$ depends on $c$ (for $p = 13$: 28 of the 169 boxes for $c = 1$, 4 for $c = 2$); it always includes the HJSW box $x_0 = -h$, $y_0 \in \{0,1\}$.

*Remark* 16 (larger boxes). In boxes larger than $2p \times 2p$ the classes have more copies; the maximum is then at least $3(p-1)$ (equal to it for $N = 2p, 2p+1$ in the cases computed and larger from $N = 2p+2$ on), and the ratio maximum/$N$ does not exceed the HJSW value $3(p-1)/2p$ in the range computed (it is not monotone in $N$; for $p = 7$ the $3p \times 3p$ box ties, $27/21 = 18/14$). Exact values (MIP with zero gap, $c = 1$, box $[-h, -h+N) \times [0, N)$): for $p = 11$ and $N = 22, 23, \ldots, 33$ the maxima are 30, 30, 31, 31, 33, 36, 36, 38, 39, 40, 42, 43; for $p = 13$, $N = 26, \ldots, 39$: 36, 36, 38, 38, 38, 38, 40, 40, 43, 45, 47, 49, 51, 52; for $p = 17$, $N = 34, \ldots, 51$: 48, 48, 49, 49, 50, 51, 52, 53, 56, 56, 58, 61, 62, 63, 65, 66, 67, 69. In the $3p \times 3p$ box the maximum is $27, 43, 52, 69, 77, 93$ for $p = 7, 11, 13, 17, 19, 23$: about $4.3(p-1)$, i.e. ratio 1.29–1.35 to the side $3p$, against $3(p-1)/2p = 1.29, 1.36, 1.38, 1.41, 1.42, 1.43$ for the HJSW box at the same primes (the value depends on $c$: $51, 52, 54$ for $p = 13$, $c = 2, 1, 3$). In the $2p \times 4p$ box the trivial bound $4(p-1)$ (two of the four collinear copies in each column) is attained for $p = 7, 11, 13$. Data: `slack/verification/windows_2p_to_3p.txt`.

| $x_0 + h \backslash y_0$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 30 | 30 | 30 | 28 | 26 | 24 | 22 | 24 | 26 | 28 | 30 |
| 1 | 28 | 28 | 28 | 30 | 28 | 26 | 24 | 26 | 28 | 30 | 28 |
| 2 | 26 | 26 | 26 | 28 | 30 | 28 | 26 | 28 | 30 | 28 | 26 |
| 3 | 24 | 24 | 24 | 26 | 28 | 30 | 28 | 30 | 28 | 26 | 24 |
| 4 | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 5 | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 6 | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 7 | 22 | 22 | 22 | 24 | 26 | 28 | 30 | 28 | 26 | 24 | 22 |
| 8 | 24 | 24 | 24 | 26 | 28 | 30 | 28 | 30 | 28 | 26 | 24 |
| 9 | 26 | 26 | 26 | 28 | 30 | 28 | 26 | 28 | 30 | 28 | 26 |
| 10 | 28 | 28 | 28 | 30 | 28 | 26 | 24 | 26 | 28 | 30 | 28 |

Table 1: $p = 11$, $c = 1$: exact maximum lawful subset of $H(1, 11)$ in the box $[x_0, x_0+22) \times [y_0, y_0+22)$, indexed by $x_0 + h \bmod p$ (rows; 0 is the HJSW position $x_0 = -h$) and $y_0 \bmod p$ (columns). Values (HiGHS MIP, witnesses through the law; they agree with the formula of Theorem 3) range from $2(p-1) + 2s = 22$ to $3(p-1) = 30$; the double-count bound $Z + 2|\mathcal{L}|$ equals 30 for 76 boxes but is attained for 20 (the others have a $(1,1)$-orbit).

## 6 Two hyperbolae: a first bound for $H(1) \cup H(-1)$

Theorems 1 and 3 concern one hyperbola. For a union of two hyperbolae in the same box the exact maxima are known only computationally (Section 7); the trivial bound is $4(p-1)$, because every column and every row of the box that is not $\equiv 0$ contains four candidate points (two lifts from each hyperbola) and a lawful set takes at most two of them. Here we prove, for $k=-1$, a bound below the trivial one by an explicit weighted line cover. Throughout this section

$$\mathcal{P}_{-1} := \big(H(1,p)\cup H(-1,p)\big)\cap G(p), \qquad |\mathcal{P}_{-1}| = 8(p-1),$$

and, for a slope-$(\pm1)$ integer line $\ell$, $|\ell|$ denotes the number of points of $\mathcal{P}_{-1}$ on $\ell$. Let $m_8(p)$ be the number of slope-$(+1)$ lines $x-y=d'$ with $|\ell|=8$.

**Theorem 17.** *For every odd prime $p$, every lawful subset of $\mathcal{P}_{-1}$ has at most $4(p-1)-4\,m_8(p)$ points.*

The proof rests on the two reflections of the box. Put $R(x,y)=(p-x,\,y)$ and $R'(x,y)=(x,\,2p-y)$. Both map $G(p)\cap\{xy\not\equiv 0\}$ onto itself (the $x$-range $[-h,3h+1]$ is symmetric about $p/2$, the $y$-range $[1,2p-1]$ about $p$; the two omitted lines $y=0$ and $x\in\{0,p\}$ carry no candidates), both map $H(1)$ onto $H(-1)$ and vice versa (since $(p-x)y\equiv -xy$ and $x(2p-y)\equiv -xy$), and both interchange slope $+1$ and slope $-1$ lines: $R\{x-y=d'\}=\{x+y=p-d'\}$, $R'\{x-y=d'\}=\{x+y=2p+d'\}$.

**Lemma 18.** *Every column $x\not\equiv 0$ of the box contains exactly four points of $\mathcal{P}_{-1}$: the two lifts of the $H(1)$-class $\kappa=(x,1/x)$ and the two lifts of the $H(-1)$-class $R'(\kappa)=(x,-1/x)$. Every row $y\not\equiv 0$ contains exactly four points: the two lifts of $\kappa=(1/y,y)\in H(1)$ and the two lifts of $R(\kappa)=(-1/y,y)\in H(-1)$.*

*Proof.* On the column $x$ the candidates are the $y\in[0,2p)$ with $xy\equiv\pm1$: two lifts of $y\equiv 1/x$ and two of $y\equiv -1/x$. Rows likewise. □

**Lemma 19** (residue groups). *Fix a residue $d\in\mathbb{F}_p$. The points of $\mathcal{P}_{-1}$ on the slope-$(+1)$ lines $x-y\equiv d$ $\pmod p$ are: the four lifts of each $H(1)$-class $\kappa$ with $x_\kappa-y_\kappa\equiv d$ (at most two classes, $\kappa$ and $\sigma\kappa$), lying on the three lines $x-y=c_\kappa-p,\ c_\kappa,\ c_\kappa+p$ with $1,2,1$ points, where $c_\kappa=x_\kappa-y_\kappa$ is computed from the base copy; and the images under $R$ of the four lifts of each $H(1)$-class $\lambda$ with $x_\lambda+y_\lambda\equiv -d$ (at most two classes, $\lambda$ and $\tau\lambda$), lying on the lines $x-y=c'_\lambda-p,\ c'_\lambda,\ c'_\lambda+p$ with $1,2,1$ points, where $c'_\lambda=-(x_\lambda+y_\lambda)$. All these centres lie in the interval $[-h-p+1,\ h-1]$ of length $2p-2$, so a residue $d$ involves at most two distinct centres, $c$ and $c+p$. In particular $|\ell|\le 8$ for every slope-$(\pm1)$ line, and $|\ell|=8$ for $\ell=\{x-y=c\}$ iff four distinct classes ($\kappa,\sigma\kappa,\lambda,\tau\lambda$; none fixed by $\sigma$ resp. $\tau$) have the common centre $c$; then the sixteen points of these four classes lie on the three lines $x-y=c-p,c,c+p$ with $4,8,4$ points, and no other point of $\mathcal{P}_{-1}$ lies on these three lines. We call this a* good residue group *of slope $+1$; good residue groups of slope $-1$ are defined symmetrically (they are the images under $R$ of the good groups of slope $+1$, so there are $m_8(p)$ of each slope).*

*Proof.* The copies of $\kappa$ are $(x_\kappa+rp,y_\kappa+sp)$, on $x-y=c_\kappa+(r-s)p$: one point each for $(r,s)=(1,0),(0,1)$ and two for $(0,0),(1,1)$. A point of $H(-1)$ on $x-y=d'$ is $R(q)$ for a point $q$ of $H(1)$ on $x+y=p-d'$; the copies of $\lambda$ lie on $x+y=x_\lambda+y_\lambda+(r+s)p$, whose $R$-images lie on $x-y=-(x_\lambda+y_\lambda)-(r+s)p+p$, i.e. centre $c'_\lambda$ with pattern $1,2,1$ (with $r+s=1$ on the centre line). Since $x_\kappa\in[-h,h]$, $y_\kappa\in[1,p-1]$, both centres lie in $[-h-p+1,h-1]$. A class fixed by $\sigma$ (resp. $\tau$) is its own partner and contributes only its own two points to the centre line, so an 8-point line needs four distinct classes with a common centre; the count of points on $c\pm p$ is then $1+1+1+1$. Only the classes with $x-y\equiv d$ (resp. $x+y\equiv -d$) have points on lines of residue $d$, so no other points occur. □

**Lemma 20.** *Let $\mathcal{G}$ be the set of all good residue groups of both slopes and let $B\subseteq\mathcal{P}_{-1}$ be the union of their points. Then (i) two distinct good groups have no point in common; (ii) $R(B)=R'(B)=B$; (iii) every row and every column of the box is either contained in $B$ or disjoint from $B$.*

*Proof.* (i) Two good groups of the same slope have different residues, hence disjoint point sets. A good group of slope $+1$ contains, from $H(1)$, a $\sigma$-pair $\{\kappa, \sigma\kappa\}$ with $c_\kappa = c_{\sigma\kappa}$ (a *shared* pair, i.e. $x_\kappa$ and $x_{\sigma\kappa} = -y_\kappa \bmod p$ have opposite signs, types $A/D$ of Lemma 7), and from $H(-1)$ the $R$-images of a shared $\tau$-pair (types $B/C$). A good group of slope $-1$ is the $R$-image of one of slope $+1$, so it contains from $H(1)$ a shared $\tau$-pair and from $H(-1)$ the $R$-images of a shared $\sigma$-pair. Since a class cannot be both $\sigma$-shared and $\tau$-shared (Lemma 7: the former means $\kappa \in A \cup D$, the latter $\kappa \in B \cup C$), an $H(1)$-class lies in at most one good group, and the same holds for $H(-1)$-classes by applying $R$. Hence the point sets are pairwise disjoint. (ii) $R$ and $R'$ map $\mathcal{P}_{-1}$ onto itself and slope-$(+1)$ lines onto slope-$(-1)$ lines, so they permute the good groups. (iii) By Lemma 18 a row consists of the lifts of $\kappa$ and of $R(\kappa)$; the four lifts of a class are in $B$ or all outside $B$ (good groups consist of whole classes), and $\kappa \subseteq B$ iff $R(\kappa) \subseteq B$ by (ii). Columns: $\kappa$ and $R'(\kappa)$. ☐

*Proof of Theorem 17.* Let $S \subseteq \mathcal{P}_{-1}$ be lawful, so $|S \cap \ell| \le 2$ for every line $\ell$. If a family $\mathcal{C}$ of lines covers $\mathcal{P}_{-1}$, then $|S| \le \sum_{\ell \in \mathcal{C}} |S \cap \ell| \le 2|\mathcal{C}|$. Take $\mathcal{C}$ = the three lines of every good residue group (both slopes: $3 \cdot 2m_8$ lines) together with every row that is disjoint from $B$. A point of $B$ lies on a line of its group (Lemma 19); a point outside $B$ lies on a row disjoint from $B$ (Lemma 20(iii)); so $\mathcal{C}$ covers. The rows disjoint from $B$ contain exactly the $8(p-1) - 32m_8$ points outside $B$, four each (Lemma 18), so there are $2(p-1) - 8m_8$ of them, and $|S| \le 2\,(6m_8 + 2(p-1) - 8m_8) = 4(p-1) - 4m_8$. ☐

(The same count with weight $\frac{1}{2}$ on the rows and on the columns disjoint from $B$ gives the same bound as a fractional cover. Theorem 17 was found independently, on the same day, by two of the agents mentioned in the disclosure; the point-by-point verification of the cover for all $19 \le p < 320$ is `slack/block_cover_km1.py`, and for $p \le 101$ `slack/km1_theorem_check.py`.)

**Proposition 21** (the number of eight-point lines)**.** *Let $X(u) \in [-h, h]$ denote the least absolute residue of $u \in \mathbb{F}_p^*$. Then $4\,m_8(p)$ is the number of pairs $(a, b) \in (\mathbb{F}_p^*)^2$ with*

$$ab^2+(a^2-1)b+a \equiv 0 \pmod p, \qquad a^2 \not\equiv -1, \quad b^2 \not\equiv 1, \qquad X(a)X(1/a) < 0, \quad X(b)X(1/b) > 0,$$

$$X(a) - X(1/a) + X(b) + X(1/b) \;=\; p\,([X(a) > 0] - [X(b) < 0]).$$

*Proof.* By Lemma 19 an eight-point line $x - y = c$ corresponds to a residue $d$ with an $H(1)$-class $\kappa = (a, 1/a)$, $a - 1/a \equiv d$, whose $\sigma$-pair is shared and non-degenerate ($\sigma\kappa \ne \kappa$, i.e. $a^2 \not\equiv -1$), and an $H(1)$-class $\lambda = (b, 1/b)$, $b + 1/b \equiv -d$, whose $\tau$-pair is shared and non-degenerate ($b^2 \not\equiv 1$), with $c_\kappa = c'_\lambda$. Eliminating $d$ from $a - 1/a \equiv -(b + 1/b)$ gives the cubic. Writing the base copies with $Y(u) = X(u) + p[X(u) < 0]$: $c_\kappa = X(a) - Y(1/a)$ and $c_{\sigma\kappa} = X(-1/a) - Y(-a) = X(a) - X(1/a) - p[X(a) > 0]$, so the $\sigma$-pair is shared iff $X(a)$ and $X(1/a)$ have opposite signs, and then $c_\kappa = X(a) - X(1/a) - p[X(a) > 0]$; likewise $e_\lambda = X(b) + Y(1/b)$, $e_{\tau\lambda} = X(1/b) + Y(b)$, the $\tau$-pair is shared iff $X(b)$, $X(1/b)$ have the same sign, and then $c'_\lambda = -e_\lambda = -X(b) - X(1/b) - p[X(b) < 0]$. The equality $c_\kappa = c'_\lambda$ is the displayed equation. Each line arises from exactly four pairs ($a \in \{a, -1/a\}$, $b \in \{b, 1/b\}$). ☐

Since $a - 1/a + b + 1/b \equiv 0$ on the cubic, the integer $L = X(a) - X(1/a) + X(b) + X(1/b) \in [-4h, 4h]$ is a multiple of $p$, i.e. $L \in \{-p, 0, p\}$; the last condition selects one of these three values.

**Proposition 22.** $m_8(p) = \frac{1}{12}\,p + O(p^{5/6}\log^3 p)$. *Consequently $\alpha(\mathcal{P}_{-1}) \le \frac{11}{3}(p-1) + O(p^{5/6}\log^3 p)$.*

*Proof.* Let $C_0 = \{(a, b) \in \mathbb{F}_p^2 : \ ab^2 + (a^2-1)b + a = 0\}$. Its only point with $a = 0$ or $b = 0$ is $(0, 0)$.

*Step 1: $C_0$ is absolutely irreducible and $|C_0(\mathbb{F}_p)| = p + O(\sqrt{p})$.* As a polynomial in $b$ over $K = \overline{\mathbb{F}_p}(a)$, $F = ab^2+(a^2-1)b+a$ has discriminant $\Delta = (a^2-1)^2-4a^2 = a^4-6a^2+1 = (a^2-3)^2-8$, whose four roots $a^2 = 3\pm2\sqrt{2}$ are distinct (their product is $1 \ne 0$ and $3\pm2\sqrt{2}$ are distinct, $p \ne 2$), so $\Delta$ is not a square in $K$ and $F$ is irreducible over $K$; being primitive in $b$ (its coefficients $a$, $a^2-1$, $a$ have no common factor), $F$ is irreducible in $\overline{\mathbb{F}_p}[a,b]$ (Gauss). Hence $C_0$ is an absolutely irreducible plane cubic and, by the Weil bound for (possibly singular) absolutely irreducible curves, $|C_0(\mathbb{F}_p)| = p + O(\sqrt{p})$ ([4, Ch. 11]; for singular curves [5]).

*Step 2: the conditions of Proposition 21 are polytope conditions on three least residues.* On $C_0$ we have $1/b = -(a - 1/a + b)$, so with $x_1 = X(a)$, $x_2 = X(1/a)$, $x_3 = X(b)$ and $T = x_1 - x_2 + x_3 \in (-3p/2, 3p/2)$ the fourth least residue is $X(1/b) = -T + p\varepsilon$ with $\varepsilon \in \{-1, 0, 1\}$ chosen so that $|-T + p\varepsilon| \leq h$, and $L = T + X(1/b) = p\varepsilon$. The conditions $X(a)X(1/a) < 0$, $X(b)X(1/b) > 0$, $L = p([x_1 > 0] - [x_3 < 0])$ therefore hold iff $(x_1, x_2, x_3)$ lies in one of the four sets

$$W_1 = \{x_1 > 0 > x_2,\ x_3 > 0,\ p/2 < T < p\}, \qquad W_2 = \{x_1 > 0 > x_2,\ x_3 < 0,\ 0 < T < p/2\},$$
$$W_3 = \{x_1 < 0 < x_2,\ x_3 > 0,\ -p/2 < T < 0\}, \quad W_4 = \{x_1 < 0 < x_2,\ x_3 < 0,\ -p < T < -p/2\}$$

(intersected with $[-h,h]^3$; the boundary hyperplanes $T \in \{0, \pm p/2, \pm p\}$, $x_i = 0$ have relative measure $O(1/p)$ and, by the box estimate of Step 3 applied to a thin box, contain $O(\sqrt{p}\log^3 p)$ of our points, so they are irrelevant below). Indeed, for $x_1 > 0 > x_2$, $x_3 > 0$ we have $T \in (0, 3p/2)$; for $T < p/2$ we get $\varepsilon = 0$ and $X(1/b) = -T < 0$ (excluded, as $X(b) > 0$), for $p/2 < T < p$ we get $\varepsilon = 1$, $X(1/b) = p - T > 0$ and $L = p$, which is the required value $p([x_1 > 0] - [x_3 < 0])$, and for $T > p$ again $X(1/b) = p - T < 0$; this gives $W_1$, and the other three cases are analogous. Each $W_i$ is a convex polytope; writing $x_i = \pm\frac{p}{2}U_i$ with $U_i$ uniform on $(0,1)$, its relative volume in $[-h,h]^3$ is $\frac{1}{8}\Pr[1 < U_1 + U_2 + U_3 < 2] = \frac{1}{8}\cdot\frac{2}{3}$ for $W_1, W_4$ and $\frac{1}{8}\Pr[0 < U_1 + U_2 - U_3 < 1] = \frac{1}{8}\cdot\frac{2}{3}$ for $W_2, W_3$ (the density of $U_1 + U_2 + U_3$ has mass $\frac{1}{6}$ on $[0,1]$ and on $[2,3]$; that of $U_1 + U_2 - U_3$ has mass $\frac{1}{6}$ on $[-1,0]$ and on $[1,2]$), in total $\frac{1}{3} + O(1/p)$.

*Step 3: equidistribution.* For $\mathbf{h} = (h_1, h_2, h_3) \in \mathbb{F}_p^3 \setminus \{0\}$ the rational function $f_{\mathbf{h}} = h_1 a + h_2 a^{-1} + h_3 b$ on $C_0$ is non-constant: if $h_3 \neq 0$ a constant value would give $b \in K$, contradicting the irreducibility of $F$ over $K$; if $h_3 = 0$ then $h_1 a + h_2/a = c$ for infinitely many $a$ forces $h_1 = h_2 = 0$. Its degree is bounded independently of $p$, so it is not of the form $g^p - g + c$ for $p$ large, and Bombieri's estimate for exponential sums along an absolutely irreducible curve [3] (see also [4, Ch. 11]) gives

$$S(\mathbf{h}) := \sum_{(a,b)\in C_0(\mathbb{F}_p),\, a\neq 0} e\Big(\frac{h_1 a + h_2 a^{-1} + h_3 b}{p}\Big) = O(\sqrt{p})$$

with an absolute implied constant. Consider the point set $\{(a, 1/a, b)/p\} \subset [0,1)^3$ (identified with the torus) indexed by the $N = |C_0(\mathbb{F}_p)| - 1 = p + O(\sqrt{p})$ points of $C_0$ with $a \neq 0$. By the Erdős–Turán–Koksma inequality [6] its discrepancy $D_N$ satisfies, for every $H \geq 1$,

$$D_N \ \leq\ C\Big(\frac{1}{H} + \frac{1}{N}\sum_{0<\|\mathbf{h}\|_\infty \leq H} \frac{|S(\mathbf{h})|}{r(\mathbf{h})}\Big), \qquad r(\mathbf{h}) = \prod_i \max(1, |h_i|),$$

and with $|S(\mathbf{h})| = O(\sqrt{p})$, $\sum_{\|\mathbf{h}\|_\infty \leq H} 1/r(\mathbf{h}) = O(\log^3 H)$ and $H = \sqrt{p}$ we get $D_N = O(p^{-1/2}\log^3 p)$. Least residues are a fixed translate of the standard residues, so axis-parallel boxes in least-residue coordinates are (unions of at most eight) boxes in the torus, and for every such box $B$ the number of points in $B$ is $N\lambda(B) + O(\sqrt{p}\log^3 p)$.

*Step 4: from boxes to the polytopes.* Fix $M \geq 1$. Each $W_i$ is $\{(x_1, x_2) \in Q_i,\ x_3 \in J_i(x_1 - x_2)\}$ with $Q_i$ a quadrant square and $J_i(s)$ an interval whose endpoints are affine in $s$ with slopes $\pm 1$. Cutting $Q_i$ into $M^2$ squares of side $p/(2M)$ and taking on each square the interval $J_i$ common to all its points, we obtain $M^2$ boxes inside $W_i$ whose union misses a subset of $W_i$ of relative measure $O(1/M)$; hence the number of our points in $W_i$ is $N\lambda(W_i) + O(M^2\sqrt{p}\log^3 p + p/M)$, and with $M = \lceil p^{1/6} \rceil$ this is $N\lambda(W_i) + O(p^{5/6}\log^3 p)$. Summing over $i$ and using $N = p + O(\sqrt{p})$, $\sum_i \lambda(W_i) = \frac{1}{3} + O(1/p)$: the number of pairs $(a,b)$ satisfying the conditions of Proposition 21, up to the $O(1)$ excluded pairs with $a^2 \equiv -1$ or $b^2 \equiv 1$, is $\frac{1}{3}p + O(p^{5/6}\log^3 p)$, so $m_8(p) = \frac{1}{12}p + O(p^{5/6}\log^3 p)$. The last statement follows from Theorem 17. $\square$

The data agree: $m_8(p)/p$ has mean 0.0825 over the 193 primes in $[200, 1500]$ (range 0.058–0.108; $1/12 = 0.0833$).

*Remark* 23. (a) The bound is attained at $p = 11$: $m_8(11) = 2$ and $\alpha(\mathcal{P}_{-1}) = 32 = 4 \cdot 10 - 8$ (exact maximum, Section 7); $m_8(13) = m_8(17) = 0$ (no gain), and $m_8(p) \geq 2$ for every prime $19 \leq p \leq 1500$ (computed), so the bound is below $4(p-1)$ from $p = 19$ on: for $k = -1$ the trivial bound $4(p-1)$ is never attained once an eight-point line exists. In terms of the total number $M_8 = 2m_8$ of eight-point lines of both slopes the bound reads $4(p-1) - 2M_8$. For $p = 19, 23, 29, 37, 47$ it coincides with the best rank-1 (LP) certificate using rows, columns and slope-($\pm 1$) lines, which is $64, 80, 96, 128, 160$; for other $p$ that LP is somewhat smaller (it also exploits 5–7-point lines with fractional weights). (b) Proposition 22 makes the bound asymptotically explicit: $\alpha(\mathcal{P}_{-1}) \leq (11/3 + o(1))(p-1)$, against the trivial $4(p-1)$ and the computed maxima $3(p-1) + O(1)$ for $p \leq 23$. The proposition quotes standard estimates (Bombieri, Erdős–Turán–Koksma) with unspecified absolute constants; the theorem itself is elementary and unconditional, and the exponent $5/6$ is not optimised. (c) The argument uses that $R$ and $R'$ are symmetries of $\mathcal{P}_{-1}$; for a general second hyperbola $H(k)$ the rows pair a class of $H(1)$ with a class of $H(k)$ that is not the image of the first under a symmetry, and the analogous cover has waste. Rank-1 certificates below $4(p-1)$ do exist for other $k$ at large $p$ (data: LP with rows, columns, $\pm 1$ lines is $3.61$–$3.89\,(p-1)$ for $k = 2$, $29 \leq p \leq 199$), but we have no clean theorem there.

## 7 Computational observations beyond one hyperbola

Nothing in this section is a theorem: it records finite computations (all witnesses verified by an independent checker of all triples; "exact" means that a solver proved optimality — SAT: unsatisfiability at the next size, MIP/CP-SAT: zero gap — and "solver-proved" upper bounds carry no independent certificate) and the picture they suggest.

**Two hyperbolae in the HJSW box.** For $\mathcal{P}_k = (H(1,p) \cup H(k,p)) \cap G(p)$ ($8(p-1)$ candidates; trivial bound $4(p-1)$, Section 6) the maximum lawful subsets are:

| $p$ | $3(p-1)$ | $k$ | max | gain | LP relax. | method / all $k$ |
|---|---|---|---|---|---|---|
| 11 | 30 | 3 | 35 | $+5$ | 36.00 | MIP, exact; over all $k$: 32–35 ($k = -1$: 32) |
| 13 | 36 | 2 | 41 | $+5$ | 47.54 | SAT (two encodings), MIP, exact; all $k$: 38–41 ($k = -1$: 40) |
| 17 | 48 | $-1$ | 54 | $+6$ | 60.15 | MIP, exact; all $k$: 49–54 |
| 19 | 54 | $-1$ | 59 | $+5$ | 63.64 | MIP, exact; $k = 2, 3$: 57 |
| 23 | 66 | $-1$ | 70–74 | $\geq +4$ | — | CP-SAT 12 h: 70 (bound 78); MIP 8 h: 70 (bound 74); $k = 2$: 68–80, $k = 3$: 69–78 |
| 29 | 84 | $-1$ | $\geq 84$ | $\geq 0$ | — | CP-SAT 8 h: 84 (bound 92); Theorem 17: $\leq 96$ |

Observations. (i) The gain over $3(p-1)$ is between $+1$ and $+6$ for *every* $k$ at $p = 11, 13, 17$ and shows no growth up to $p = 19$; the exact value at $p = 23$ is not settled. As an explicit falsifiable conjecture: $\alpha(\mathcal{P}_k) \leq 3(p-1) + 6$ for all $p \geq 11$ and all $k \notin \{0, 1\}$ (consistent with all exact values; at $p = 23$ the range 70–74 still allows $+8$; a certified witness with gain $\geq 7$ would refute it). (ii) The LP relaxation with all rich lines is far from the truth (column "LP"; $\approx 4(p-1) - \varepsilon$), because for two hyperbolae the fractional 2-factor $x \equiv \frac{1}{2}$ is almost feasible; certificates need lines with at least five candidates (Section 6). Rank-1 certificates using only rows, columns and slope-($\pm 1$) lines give $\approx 3.45\,(p-1)$ for $k = -1$ ($19 \leq p \leq 199$) and $3.6$–$3.9\,(p-1)$ for $k = 2, 3$ ($p \leq 199$), and the integer optimum of that restricted system equals its LP value up to 2 ($k = -1$, $p \leq 59$; `slack/verification/ip1_km1.log`): so no local inequality of any rank on those lines can certify less; the remaining loss down to $3(p-1) + O(1)$ is carried by lines of unboundedly many slopes. (iii) Structure of the exact maxima: they are balanced between the two hyperbolae ($23 + 18$ at $p = 13$, $30 + 29$ at $p = 19$), use one or two copies of almost every class, and the two copies of a class are usually in one column or one row (at $p = 19$ the optimum consists of 23 such vertical pairs and 13 single copies; at $p = 17$ some classes use three copies). For $k = -1$ the points fall into $(p-1)/2$ "orbit blocks" (the 16 points of $\{\kappa, \nu\kappa, R\kappa, R'\kappa\}$ form a full $4 \times 4$ grid on its rows and columns), the trivial bound is 8 per block, and the optima have $\approx 6$ per block on average with the deficits spread over most blocks; pairs and triples of blocks essentially never conflict, so the loss is a global effect. Restricted models (only vertical/horizontal copy pairs; or $y$-periodic sets) reach $3(p-1) + O(1)$ but not the optimum. In the "vertical-pair" model for $k = -1$ (every class contributes both lifts in one of its two columns; $2^{p-1}$ orientations, each a 2-factor of the row/column structure) the number $T$ of collinear triples is a *quadratic* form $E + \varepsilon^{\mathsf{T}} C \varepsilon$ in the orientation signs (odd degrees vanish by the reflection $R'$), so $\min T \ge E + (p-1)\lambda_{\min}(C)$; numerically this spectral bound is $\ge 1.9\,(p-1)$ for $11 \le p \le 19$ (true minima $24, 28, 42, 48$; 80 at $p = 23$ and 136 at $p = 29$ against spectral bounds 64 and 129) and $\ge 3.6\,(p-1)$ for $41 \le p \le 71$ — every orientation carries linearly many collinear triples, with a certificate that is not a union bound; turning it into a theorem would need the growth of $E$ ($E/(p-1) \approx 3.6 \log p - 4.4$ up to $p = 199$) and a Wigner-type edge estimate for the dense $\pm$-matrix $C$ (numerically $\lambda_{\min} \approx -2.4\sqrt{\operatorname{tr} C^2/(p-1)} \sim -\sqrt{\log p}$, with semicircle-like normalised moments), both arithmetic statements about $\mathcal{P}_{-1}$ (`slack/vp_quadratic.py`, `slack/vp_traces.py`).

**Other unions.** A parabola $y = x^2 + k$ together with $H(1)$ in the same box: exact maxima $34, 39, 51, 58$ for $p = 11, 13, 17, 19$ ($+4, +3, +3, +4$; MIP). Three hyperbolae $H(1) \cup H(2) \cup H(3)$ ($12(p-1)$ candidates): 37 at $p = 11$ (exact, MIP; $+7$), between 43 and 48 at $p = 13$ and between 56 and 63 at $p = 17$ (CP-SAT 4 h, not closed). Two hyperbolae with different moduli, $H(1,p) \cup H(1,q)$ in the $2p \times 2p$ box (each with its own HJSW shift): the exact maxima (MIP) exceed $3(p-1)$ by 1, 2 or 3 in all eleven pairs computed, $7 \le p < q \le 29$ (e.g. 32 vs 30 for $(11,13)$, 68 vs 66 for $(23,29)$), and the number of collinear triples with points on both hyperbolae grows like $N \log N$ in the range computed ($N \le 586$; ratio to $N \ln N$ near 11), as for random sets of the same density. Larger boxes for one hyperbola: Remark 16.

**A heuristic.** Fix the HJSW set $S_2$ of $H(1,p)$ and call a point $q$ of $H(k,p) \cap G(p)$ blocked if some line through $q$ carries two points of $S_2$. The average number of blocking pairs per point grows like $\log p$ (about 3.1 at $p = 11$ and 7 at $p = 101$ for $k \in \{2, 3, -1\}$), and the number of unblocked points is 0–2 for every $p \le 101$ tested (for $k = 2$ at $p = 5$ and a few other $(p,k)$ it is 4); this depends on the chosen maximum set of $H(1)$ (over the $9^s$ maximum sets the number of unblocked points varies between 0 and 2 at $p = 13$, $k = 2$) and does not lower-bound the deletion cost, but it explains why the second hyperbola contributes little: freeing a point of $H(k)$ costs deletions in $H(1)$. Whether the gain is $O(1)$, $O(\log p)$ or $O(\sqrt{p})$ cannot be decided from the data ($+5, +5, +6, +5$); Theorem 17 shows that for $k = -1$ the gain over $3(p-1)$ is at most $(\frac{2}{3} + o(1))(p-1)$, and no more is proved. (Definitions and data: `slack/`, `docs/research/slack_experiments.md`, `docs/research/pair_bound_notes.md`.)

## 8 Independent computer verification

The theorem has a short self-contained proof; the following check is an independent sanity test of the bookkeeping (the classification of the rich lines, the exceptional classes and the count of the maximum sets), performed by a program that trusts no part of the proof. The script `slack/hjsw_window_check.py` of the repository recomputes, from the point set alone: all lines with at least three points of $\mathcal{P}$ (found by brute force over all pairs of points; all have slope $\pm 1$, three or four points, and lie inside one $V$-orbit); the identity $2|\mathcal{L}| + Z = 3(p-1)$; the exact maximum by exhaustive search inside each orbit (at most $2^{16}$ subsets), assembling a maximum witness which is passed through the independent verifier `saturation.the_law` in grid coordinates; and the number of maximum sets, compared with $9^s$. All checks pass for every prime $3 \le p \le 101$ (all $c \in \mathbb{F}_p^*$ for $p \le 31$; $c \in \{1, 2, 3, -1, -2, (p+1)/2\}$ for larger $p$); log: `slack/verification/hjsw_window_check_all.log`. For Theorem 3 the script `slack/hjsw_boxes_check.py` (log `slack/verification/hjsw_boxes_check.log`) checks, for every box $(x_0, y_0 \bmod p)$ and every $c$: the formula $Z + 2|\mathcal{L}| = 2(p-1) + 2s + 2\sum_O (e_O + f_O)$ against brute-force line enumeration (random boxes, $p \le 19$); $Z + 2|\mathcal{L}| \le 3(p-1)$ for all primes $p \le 61$ (with equality for some boxes); the orbit lemma orbit by orbit ($(e_O, f_O) \in \{(0,0), (0,2), (1,1), (2,0)\}$ only) for all $p \le 31$; every identity used in the proof of Lemma 10 ($\theta = c_a + c_b - k$, $\eta = c_b - c_a$, the four split conditions, parity, and "$F_1 \Rightarrow \neg(E_1 \wedge E_2)$") on random instances up to $p = 199$; the maxima and multiplicities of Lemma 11 orbit by orbit (exhaustive search over the $2^{16}$ subsets) for all $p \le 13$, all $c$ and all boxes; and the formula $12n_2 + 10n_1 + 8n_0 + 6s$ against MIP optima on random boxes with $p \le 31$. The exact maxima of Table 1 are MIP optima (HiGHS) with witnesses through the law. The values agree with the exact maxima obtained earlier in the project by SAT solving (kissat/CaDiCaL, two encodings, witnesses through the law): $18, 30, 36, 48, 54, 66$ for $p = 7, 11, 13, 17, 19, 23$. Two external LLM-assisted audits (ChatGPT deep-research reports commissioned by the author of record on v0.1 and on v0.5 of this note; kept in `docs/reviews/`) re-derived the point sets and the rich lines with clean-room implementations and confirmed $3(p-1)$, $9^s$, the LP value and the no-four values for all $c$ with $p \le 31$ (samples for larger $p$), all box patterns and gadget counts through $p = 13$, and the shifted-box table for $p = 11$; the second audit also identified a gap in the earlier presentation of Lemma 11 (the $(1,1)$ count), which is now proved above.

Repository (public): `https://github.com/iwasborninbali/saturation`.
This version of the note is the immutable tag `hjsw-note-v1.0`: file `paper/hjsw_window.tex`; scripts in `slack/`: `hjsw_window_check.py`, `hjsw_boxes_check.py`, `gadget11_check.py`, `km1_theorem_check.py`, `block_cover_km1.py`, `km1_lines.py`; logs in `slack/verification/`; checksums and the list of tags in `docs/RELEASES.md`.

| $p$ | $c$ | $\lvert\mathcal{P}\rvert$ | $\lvert\mathcal{L}\rvert$ | $Z$ | max | #max sets |
|---|---|---|---|---|---|---|
| 7 | 1 | 24 | 8 | 2 | 18 | 9 |
| 11 | 1 | 40 | 14 | 2 | 30 | 9 |
| 13 | 1 | 48 | 16 | 4 | 36 | 81 |
| 13 | 2 | 48 | 18 | 0 | 36 | 1 |
| 17 | 1 | 64 | 22 | 4 | 48 | 81 |
| 19 | 1 | 72 | 26 | 2 | 54 | 9 |
| 23 | 1 | 88 | 32 | 2 | 66 | 9 |
| 101 | 1 | 400 | 148 | 4 | 300 | 81 |

Table 2: Sample values from the verification log ($|\mathcal{L}| = \frac{3}{2}(p-1) - s$, $Z = 2s$).

**AI/tool-use disclosure and authorship statement.** The theorems, their proofs, the verification programs and the text of this note were produced by autonomous large-language-model agents (Anthropic Claude, model *Claude Fable 5*, August 2026) working in a shared repository: one agent found and proved Theorems 1 and 3 and wrote the first version of the note; Theorem 17 was found independently, on the same day, by two of the agents; the computational setup and the observations of Section 7 come from all three; the note was revised by the agents after two external LLM-assisted audits commissioned by the author of record. The author of record posed the question, directed and organised the work, commissioned the external audits, decided what to claim and publish, and takes responsibility for the content; the extent of his independent verification of the proofs is recorded in `docs/HUMAN_AUDIT.md` of the repository and will be stated precisely in any submitted version. The AI systems are not authors. All code, logs and the correspondence of the agents are public in the repository.

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